Writing NPU quant kernels: SIMT vs PTO vs ASC

Code size and simulated performance of three tilelang vector frontends implementing one group-wise quantization algorithm.

- 22 kernel variants, each implemented in ASC, PTO and SIMT plus a PyTorch reference: 88 kernels, all passing `python -m harness.check`
- msprof-simulated VF cycles for all 66 NPU kernels
- tilelang 0.1.15, ptoas vmi 0.1.9, CANN 9.2.0-beta.2, SoC `Ascend950PR_9599`
- Single core, `T.Kernel(1)`. Only the `T.SimdVF` / `T.SimtVF` region differs between the three implementations of a variant; tiling, DMA and buffer allocation are identical.

tilelang frontend comparison

Group-wise quantization, in PyTorch:

```
grouped = x.float().view(m, k // group_size, group_size)
amax = grouped.abs().amax(dim=-1).clamp(min=E4M3_CLAMP_MIN)
sf = amax / E4M3_MAX
q = (grouped * (E4M3_MAX / amax).unsqueeze(-1)).view(m, k).to(torch.float8_e4m3fn)
```

frontend	abstraction level	smallest unit of data
ASC (`T.simd`, `S`)	named hardware instruction and distribution mode	one 64-lane vector register
PTO/VMI (`T.vmi`, `V`)	operation with width and segment count as arguments	one logical vector, any width
SIMT (`T.SimtVF`, `T.Parallel`)	per-element expression	one element, held by one thread

`T.SimdVF` / `T.SimtVF` regions range from **6 to 57 lines** across the 22 variants. msprof-simulated VF cycles for the same variant differ by **10-100x between the SIMT and the SIMD implementations**, and by up to 1.6x between the two SIMD frontends.

ASC vs PTO: fewer operations, comparable cycles

PTO expresses the same 22 kernels in **74% of ASC's vector operations** and **89% of its source lines**. Simulated cycles do not follow:

PTO relative to ASC	
vector operations written	74%
`T.SimdVF` source lines	89%
simulated VF cycles	**0.92x mean**, range 0.64x-1.13x

- PTO uses fewer cycles on **9 of 22** variants, more on 6, and is within 2% on 7
- On 4 of those 9 it writes *more* operations than ASC
- Operations written and cycles executed correlate at **Pearson r = -0.09** (n = 22): across this set, source size carries no information about cycle count

Both SIMD frontends are 3-100x faster than SIMT on every variant.

Benchmark setup (and caveats resolved)

Cycle measurement. Sum the `cycles` column of `core0.veccore0_instr_exe.csv` over rows whose `instr` is a VF launch (`VF` for SIMD, `VF_SIMT` for SIMT). The filter keys on `instr`, not `pipe`: every VF row sits on pipe `PUSHQ`, which `MOVEMASK` and `PUSH_PB` also use.

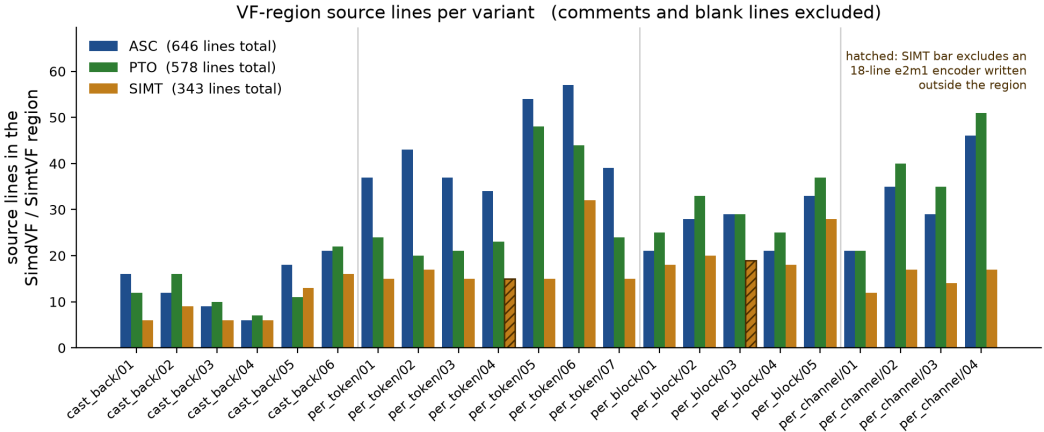
Correctness. Each kernel is compared against `harness/oracle.py`, with an assertion that the result was read back from the device. Cycles are not reported for a kernel that computed a wrong result.

Equal work. VF launch counts agree across the three frontends on every variant except `per_token/05` — ASC and PTO issue 34, SIMT 32, the difference being the gather index SIMT does not build.

Two measurement errors found and corrected. `per_token/06` and `per_block/05` allocated their input with `torch.zeros(device=...)`, an aten fill *kernel*; msprof profiles the first operator, so all three frontends had measured that fill. Both are now driven by `tools/vf_drive.py`, and `tools/vf_trace_audit.py` checks all 66 instruction logs for the arithmetic a quant kernel must contain.

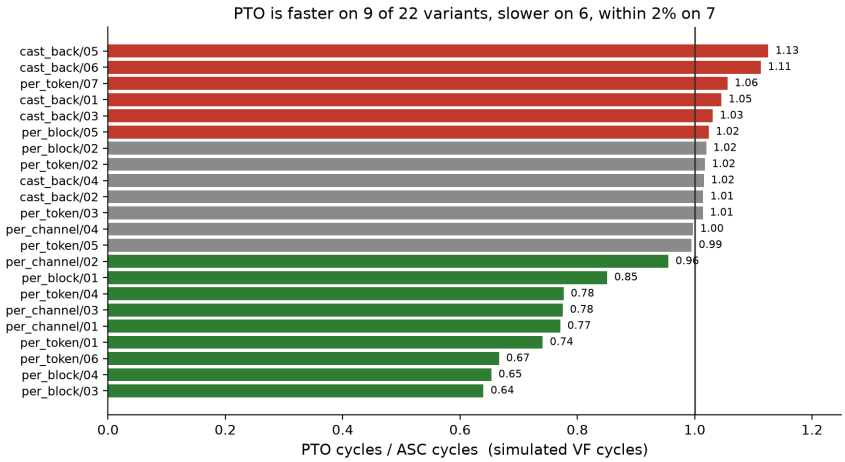
Reading the code fragments. Each is cut verbatim from a kernel the harness passes, and re-checked on every build. Within a variant the three fragments cover the same step; the SIMD fragments' enclosing `for ... in T.serial(...)` tile loop is elided and the SIMT fragment's `T.Parallel` header kept, because that header is what replaces it.

Source size: VF-region lines, all three frontends

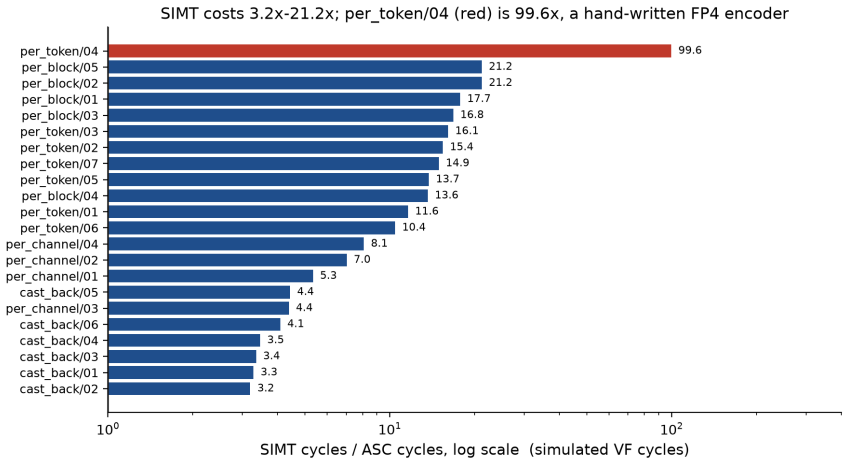


Totals **646 / 578 / 343** lines — PTO 89% of ASC, SIMT 53%. PTO is *longer* than ASC on 11 of the 22: its saving is concentrated in the segmented-reduction variants (A2) and spent again on `size=` and mask operands elsewhere (B3). SIMT is shorter on 21 of 22 and ties on the other.

PTO vs ASC: simulated cycles per variant



SIMT vs ASC: simulated cycles per variant



`per_token/04` is separated because its cost is a convert the SIMT surface cannot reach, not a property of the thread model (D1). The other 21 variants span 3.20x-21.18x.

Walkthrough: nine variants, four kinds of layout mismatch

The algorithm is the same in all 22 variants. What changes is where its quantities sit relative to the 64- or 128-lane vector register.

	variant	code pattern	what it decides
A reduce	`per_channel/01`	one channel per lane	nothing to simplify
	`per_token/01`	4 groups per register	PTO's clearest win
	`per_block/01`	32x32 tile to a scalar	counts mispredict cycles
B scale	`cast_back/01`	one scalar across lanes	SIMT's first real cut
	`per_token/02`	exponent, any width	PTO's width argument
	`cast_back/02`	nothing to factor out	PTO's operands cost lines
C output	`per_channel/02`	interleave two registers	SIMT drops an instruction
	`per_token/05`	transposed store = gather	biggest cut, and its price
D convert	`per_token/04`	float32 -> e2m1	SIMT's 99.6x

The other 13 variants compose these patterns and are measured in the tables at the end. Long form: companion report, sections 2-8.

A1 `per_channel/01`: the reduction axis is already the lane axis

One scale per channel: the reduction runs down the token axis, so a channel stays in one lane and no frontend performs a cross-lane operation.

ASC, accumulating across the 32 rows:

```
acc = S.alloc_local((1,), T.float32)
# seeding with the clamp floor makes the clamp free
acc[0] = S.vdup(E4M3_CLAMP_MIN, T.float32)
for row in T.serial(group_tokens):
    v = S.vabs(S.vcvt(S.vld(x_ub[row, col], dist="UNPK_B16"),
                      T.float32))
    # lane c holds the max over tokens seen so far
    acc[0] = S.vmax(acc[0], v)
S.vsts(sf_ub[col], S.vdiv(acc[0], qmax))
S.vsts(inv_ub[col], S.vdiv(qmax, acc[0]))
```

PTO — the same, with mask and width operands:

```
acc = V.alloc_local((1,), V.vreg(LANES, T.float32))
# seeding with the clamp floor makes the clamp free
acc[0] = V.vbrc(T.float32(E4M3_CLAMP_MIN), size=LANES)
for row in T.serial(group_tokens):
    v = V.vabs(V.vcvt(V.vload(x_ub[row, col], size=LANES),
                      "float32"), mask)
    # lane c holds the max over tokens seen so far
    acc[0] = V.vmax(acc[0], v, mask)
V.vstore(V.vdiv(acc[0], qmax, mask), sf_ub[col])
V.vstore(V.vdiv(qmax, acc[0], mask), inv_ub[col])
```

A1 per_channel/01: equal source, unequal cycles

SIMT assigns one thread per channel and loops over the 32 rows:

```
for c in T.Parallel(hidden):
    amax = T.alloc_var(T.float32)
    amax = E4M3_CLAMP_MIN
    for row in T.serial(group_tokens):
        amax = T.max(amax,
                     T.abs(T.Cast("float32", x_ub[row, c])))
    sf_ub[c] = amax / E4M3_MAX
    inv_ub[c] = E4M3_MAX / amax
```

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
per_channel/01	502	387 (0.77x)	2,677 (5.33x)	21 / 21 / 12

17 and 16 vector operations, 21 lines each: segment counts and width arguments have nothing to apply to. Cycles differ anyway — 64 lanes against 128, so the elided tile loop runs twice, then once.

> ***A1*** *— where the data lines up with the lanes, PTO has nothing to factor out and SIMT nothing to remove. The 0.77x is register width, which no source count can see; it recurs, larger, at A3.*

A2 per_token/01: a segmented reduction is PTO's clearest win

Each 32-channel group needs its own maximum. An ASC vector reduction produces one value per register, so 4 groups require 4 masked reductions and 4 single-element stores:

```
a0 = S.vabs(S.vcvt(S.vld(x_ub[col], dist="UNPK_B16"), T.float32))
a1 = S.vabs(S.vcvt(S.vld(x_ub[col + LANES], dist="UNPK_B16"),
                   T.float32))
# Each register holds two groups, so reduce half at a time.
S.vsts(amax_ub[group], S.vcmax(a0, mask_low), dist="ONEPT_B32")
S.vsts(amax_ub[group + 1], S.vcmax(a0, mask_high),
       dist="ONEPT_B32")
S.vsts(amax_ub[group + 2], S.vcmax(a1, mask_low),
       dist="ONEPT_B32")
S.vsts(amax_ub[group + 3], S.vcmax(a1, mask_high),
       dist="ONEPT_B32")
```

PTO declares the segment count on one reduction:

```
x = V.vcvt(V.vload(x_ub[col], size=PAIR), "float32")
V.vstore(V.vcmax(V.vabs(x, mask), mask, group=groups_per_pair),
         amax_ub[group])
```

14 vector operations become 5. The matching broadcast on the apply pass collapses from 4 loads plus 2 selects to 1 load.

A2 per_token/01: SIMT, and the measured result

SIMT assigns one thread per group:

```
for g in T.Parallel(num_groups):
    amax_l[g] = E4M3_CLAMP_MIN
    for j in T.serial(group_size):
        v = T.Cast("float32", x_ub[g * group_size + j])
        amax_l[g] = T.max(amax_l[g], T.abs(v))
    inv_l[g] = E4M3_MAX / amax_l[g]
    sf_l[g] = amax_l[g] / E4M3_MAX
```

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
per_token/01	6,176	4,576 (0.74x)	71,606 (11.59x)	37 / 24 / 15

`T.Parallel(num_groups)` has extent 4 at `K=128`, `group_size=32`, so the SIMT reduction uses 4 of the 128 declared threads.

> ***A2*** *— the one pattern where PTO is both shorter and faster: 19 operations against ASC's 39 at 0.74x the cycles. Operation count and cycle count favour PTO together on 6 of the 22 variants. SIMT is shorter again (15 lines) at 11.59x.*

A3 `per_block`: the same tile, half as many steps

One scalar per 32x32 tile. **ASC** takes 16 steps of 64 lanes:

```
for chunk in T.serial(num_chunks):
    v = S.vabs(S.vcvt(S.vld(flat_x[chunk * LANES],
                            dist="UNPK_B16"), T.float32))
    S.vsts(run_ub[chunk], S.vcmax(v, mask_all),
           dist="ONEPT_B32")
```

PTO reaches 128 float32 lanes, so the tile takes 8 steps of 128:

```
for chunk in T.serial(num_chunks):
    v = V.vabs(V.vcvt(V.vload(flat_x[chunk * LANES],
                              size=LANES), "float32"), mask)
    V.vstore(V.vcmax(v, mask, group=1), run_ub[chunk])
```

per_block	/01	/02	/03	/04	/05
operations, ASC -> PTO	21->22	27->30	28->26	21->22	29->31
cycles, PTO/ASC	**0.85x**	1.02x	**0.64x**	**0.65x**	1.02x

PTO writes more operations on 4 of these 5 and runs fewer cycles on 2 of those 4.

A3 per_block/01: SIMT has no tile loop, and reduces in two levels

A thread cannot see another thread's accumulator, so the tile reduction becomes a tree. Level 1 gives each of the 32 rows a thread:

```
# level 1: one thread per row of the tile
for t in T.Parallel(bm):
    amax = T.alloc_var(T.float32)
    amax = E4M3_CLAMP_MIN
    for j in T.serial(bk):
        amax = T.max(amax,
                     T.abs(T.Cast("float32", x_ub[t, j])))
    row_ub[t] = amax
```

Level 2 is a second region, `for _z in T.Parallel(1)`, folding the 32 partials to the scalar — one thread working while 127 wait.

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
per_block/01	1,086	924 (0.85x)	19,276 (17.75x)	21 / 25 / 18

> ***A3*** *— more operations per step and half as many steps executed: a static count cannot observe a trip count. SIMT has neither — no tile loop and no vector width — and pays 17.75x for the serial fold.*

B1 `cast_back/01`: broadcasting one scalar across a lane range

64 consecutive channels span two scale groups: each lane needs one of two scalars. **ASC** — two broadcast loads and a lane-mask select:

```
lo = S.vld(sf_ub[group], dist="BRC_B32")
hi = S.vld(sf_ub[group + 1], dist="BRC_B32")
scale = S.vsel(lo, hi, mask_low)

scaled = S.vmul(values, scale)
```

PTO — the segment structure declared on the load:

```
scale = V.vload(sf_ub[group], size=LANES, stride=1,
                dist_mode="brc", group=groups_per_strip)

scaled = V.vmul(values, scale, mask)
```

SIMT — an index expression:

```
for i in T.Parallel(hidden):
    o_l[i] = q_l[i] * sf_ub[i // group_size]
```

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
`cast_back/01`	2,105	2,201 (1.05x)	6,908 (3.28x)	16 / 12 / 6

B1 cast_back/01: output dtype selection, and the result

The variant also switches output dtype. **ASC** picks a different store instruction; the wider dtype takes the simpler one:

```
if out_dtype == "bfloat16":
    S.vsts(out_ub[col], S.vcvt(scaled, T.bfloat16),
           dist="PK_B32")      # 64 lanes -> 64 x 2 bytes
else:
    S.vsts(out_ub[col], scaled,
           dist="NORM_B32")    # 64 lanes -> 64 x 4 bytes
```

PTO infers the packing from the destination buffer:

```
if out_dtype == "bfloat16":
    V.vstore(V.vcvt(scaled, "bfloat16"), out_ub[col])
else:
    V.vstore(scaled, out_ub[col])
```

SIMT emits no dtype branch at all: `T.copy` converts between buffers.

> ***B1*** *— SIMT's source is materially shorter for the first time (16 lines to 6): per-element indexing replaces explicit lane-mask construction, at 3.28x, the 2nd lowest SIMT ratio of 22.*

B2 `per_token/02`: lane count as a parameter

The scale is rounded up to a power of two by rewriting the float32 exponent field. **ASC** fixes the lane count in every type name (`"uint32x64"`):

```
one = S.vdup(1, T.uint32)
bits = T.reinterpret(S.vmuls(clamped, inv_qmax), "uint32x64")
# ((bits - 1) >> 23) + 1  ==  ceil(log2(v)) + 127
biased = S.vadds(S.vshrs(S.vsub(bits, one), 23), 1)
sf = T.reinterpret(S.vshls(biased, 23), "float32x64")
# 254 - biased == 127 - ceil_exp: the negated exponent
inv = T.reinterpret(
    S.vshls(S.vsub(S.vdup(254, T.uint32), biased), 23),
    "float32x64")
```

PTO takes the width as an argument, so one `@T.macro` body — `compute_scale(amax, lanes)` — serves 4, 64, 128 or 256 lanes:

```
one = V.vbrc(T.uint32(1), size=lanes)
shift = V.vbrc(T.uint32(23), size=lanes)
bits = V.vinterpret_cast(
    V.vmul(clamped, V.vbrc(T.float32(inv_qmax), size=lanes), mask), "uint32")
# ((bits - 1) >> 23) + 1  ==  ceil(log2(v)) + 127
biased = V.vadd(V.vshr(V.vsub(bits, one), shift), one)
sf = V.vinterpret_cast(V.vshl(biased, shift), "float32")
# 254 - biased == 127 - ceil_exp: the negated exponent, still biased.
inv = V.vinterpret_cast(
    V.vshl(V.vsub(V.vbrc(T.uint32(254), size=lanes), biased), shift), "float32")
```

B2 `per_token/02`: the same arithmetic on scalars

SIMT has no vector width to carry. Its first six lines are A2's reduction unchanged; the scale arithmetic is the last four:

```
for g in T.Parallel(num_groups):
    amax = T.alloc_var(T.float32)
    amax = E4M3_CLAMP_MIN
    for j in T.serial(group_size):
        v = T.Cast("float32", x_ub[g * group_size + j])
        amax = T.max(amax, T.abs(v))

    bits = T.reinterpret("int32", amax / E4M3_MAX)
    biased = ((bits - 1) >> 23) + 1
    sf_l[g] = T.reinterpret("float32", biased << 23)
    inv_l[g] = T.reinterpret("float32", (254 - biased) << 23)
```

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
`per_token/02`	4,608	4,688 (1.02x)	71,008 (15.41x)	43 / 20 / 17

> ***B2*** *— 45 operations against 14 is the largest operation-count reduction in the set, at a 1.02x cycle ratio. PTO's gain here is a definition that is written once instead of per width; it is not a speed gain, and reporting only the operation count would have presented it as one.*

B3 `cast_back/02`: where PTO's explicitness costs source length

Decoding a UE8M0 exponent byte. **ASC** names the distribution mode:

```
packed = S.vld(sf_ub[strip], dist="BRC_B16")
bits = S.vand(S.vshl(T.reinterpret(packed, "uint32x64"),
                     sf_shift), exp_mask)
# An exponent field with a zero mantissa *is* 2^(e-127).
scale = T.reinterpret(bits, "float32x64")
```

PTO states width, stride, group count and mask:

```
packed = V.vload(sf_ub[strip], size=LANES * 2, stride=1,
                 dist_mode="brc", group=1)
bits = V.vand(V.vshl(V.vinterpret_cast(packed, "uint32"),
                     sf_shift), exp_mask)
scale = V.vinterpret_cast(bits, "float32")
```

SIMT shifts by a per-element amount, which VMI cannot express — `V.vshrs` takes a scalar:

```
for i in T.Parallel(hidden):
    g = i // group_size
    word = T.Cast("int32", sf_ub[g // PACK_FACTOR])
    byte = (word >> (8 * (g % PACK_FACTOR))) & 0xFF
    o_l[i] = q_l[i] * T.reinterpret("float32", byte << 23)
```

> ***B3*** *— the counter-example to B2. With nothing to factor out, operations tie at 12 and PTO runs 16 lines against 12; SIMT writes 9 at 3.20x, the tier's lowest.*

C1 per_channel/02: M-packed scales require a lane interleave

Output byte 2c holds m-group 2i's exponent for channel c, byte 2c+1 holds m-group 2i+1's. On a vector unit this is an interleave instruction.

ASC:

```
a = S.vld(sf_rows_ub[0, col], dist="NORM_B8")
b = S.vld(sf_rows_ub[1, col], dist="NORM_B8")
# A uint8 register is 256 lanes, so `lo` already holds the
# interleave of the first 128 elements of each input --
# exactly the 256 output bytes for these 128 channels.
# `hi` interleaves the padding and is discarded.
lo, _ = S.vintlv(a, b)
S.vsts(packed_ub[col * PACK_FACTOR], lo, dist="NORM_B8")
```

PTO — same instruction, with the vector width given explicitly:

```
a = V.vload(sf_rows_ub[0, col], size=BYTE_LANES * 2)
b = V.vload(sf_rows_ub[1, col], size=BYTE_LANES * 2)
# A uint8 register is 256 lanes, so `lo` already holds the
# interleave of the first 128 elements of each input --
# exactly the 256 output bytes for these 128 channels.
# `hi` interleaves the padding and is discarded.
lo, _ = V.vintlv(a, b, byte_mask)
V.vstore(lo, packed_ub[col * PACK_FACTOR])
```

C1 per_channel/02: SIMT replaces the interleave with two scalar stores

```
for c in T.Parallel(hidden):
    packed_ub[PACK_FACTOR * c] = sf_rows_ub[0, c]
    packed_ub[PACK_FACTOR * c + 1] = sf_rows_ub[1, c]
```

A uint8 vector register holds 256 lanes, so the loads above read 128 bytes past the 128 meaningful ones. Both SIMD frontends therefore pad the scratch buffer by a whole tile:

```
sf_rows_ub = T.alloc_shared((PACK_FACTOR, hidden + BYTE_LANES), T.uint8)
```

```
sf_rows_ub = T.alloc_shared((PACK_FACTOR, hidden + BYTE_LANES), T.uint8)
```

A scalar store cannot overrun, so SIMT allocates the exact size:

```
sf_rows_ub = T.alloc_shared((PACK_FACTOR, hidden), T.uint8)
```

C1 `per_channel/02`: the measured cost of removing the interleave

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
`per_channel/02`	849	811 (0.96x)	5,964 (7.02x)	35 / 40 / 17

- SIMT removes one instruction and one padding requirement, using 17 lines against ASC's 35, at **7.02x** the cycles
- PTO is **5 lines longer** than ASC (40 vs 35) at 0.96x the cycles: the explicit `size=` and mask operands cost source length, not time

> ***C1*** *— SIMT's largest structural simplification so far: it deletes an instruction and a correctness rule about register width. Both deletions are real; 7.02x is what they cost.*

C2 `per_token/05`: column-major scales require a register gather

Lane assignment inside a vector register is fixed, so transposing the scale array means computing, per lane, which source element that lane receives.

ASC builds the index vector from the lane id:

```
lane = S.vci(0, T.int32)
token_of_lane = S.vand(T.reinterpret(lane, "uint32x64"),
                       S.vdup(token_block - 1, T.uint32))
group_of_lane = T.reinterpret(S.vshrs(lane, log2_block), "uint32x64")
S.vsts(idx_ub[0],
       S.vadd(S.vmuls(token_of_lane, SF_STRIDE), group_of_lane),
       dist="NORM_B32")
```

. . . and gathers through it:

```
S.vsts(sf_out_ub[base, 0],
       S.vgather2(sf_dense_ub[0, base], idx), dist="NORM_B32")
```

C2 `per_token/05`: PTO builds the same index; SIMT changes one store

PTO builds the same index vector. VMI has no scalar-operand `vmul`, so the stride is broadcast first — 10 lines against ASC's 7 — and the gather needs its mask:

```
lane = V.vci(T.int32(0), size=LANES)
mask64 = V.create_mask(LANES, size=LANES)
token_of_lane = V.vand(V.vinterpret_cast(lane, "uint32"),
                       V.vbrc(T.uint32(token_block - 1), size=LANES))
group_of_lane = V.vinterpret_cast(
    V.vshrs(lane, log2_block, mask64), "uint32")
# No scalar-operand vmul in VMI: the stride must be a vector.
stride_v = V.vbrc(T.uint32(SF_STRIDE), size=LANES)
V.vstore(V.vadd(V.vmul(token_of_lane, stride_v, mask64),
                group_of_lane), idx_ub[0])
```

```
V.vstore(V.vgather(sf_dense_ub[0, base], idx, gmask),
         sf_out_ub[base, 0])
```

SIMT stores the scale at the transposed index. The fragment is one statement because the rest of its `T.Parallel(num_groups)` body is B2's, unchanged: the column-major configuration changes which index this store uses and nothing else. There is no index vector and no gather to show.

```
# The whole column-major config: a different index.
sf_out_ub[g, sub] = T.reinterpret("float32", biased << 23)
```

C2 `per_token/05`: the sharpest source-size / cycle trade in the set

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
`per_token/05`	5,477	5,448 (0.99x)	75,281 (13.74x)	54 / 48 / 15

- 15 SIMT lines against ASC's 54 is the largest source-size reduction among the 22 variants
- SIMT costs **13.74x** the cycles, the 14th lowest SIMT ratio of 22
- ASC and PTO are within 1% of each other: a gather is a single hardware operation, so VMI has no segment structure or width to collapse here

> ***C2*** *— one line of SIMT replaces nine of index arithmetic, and the SIMD frontends tie because the hardware already provides the operation. Where the work is index arithmetic, SIMT is the shortest by a wide margin and the slowest by one too.*

D1 per_token/04: no FP4 convert is reachable from a SimtVF region

The hardware convert exists and all three frontends run on it. ASC and PTO name it — `S.vcvt(v, T.float4_e2m1fn)`, `V.vcvt(v, "float4_e2m1fn")` — so their work is limited to pre-rounding to bfloat16 with a sticky bit. Nothing in the SIMT surface reaches it: inside a `T.SimtVF` region the arithmetic is plain operators on scalars, and `T.copy` into an FP4 buffer does not compile (`harness/probe/simt_limits.py`).

ASC pairs two registers to deinterleave them, then converts:

```
low, high = S.vdintlv(T.reinterpret(q0, "uint16x128"),
                      T.reinterpret(q1, "uint16x128"))
odd = S.vor(high, S.vmin(low, one_u16))
S.vsts(q_ub[col],
       S.vcvt(T.reinterpret(odd, "bfloat16x128"),
              T.float4_e2m1fn), dist="PK4_B32")
```

PTO uses `V.vunzip`, which splits one value instead of pairing two. It is a `@T.macro`, so the call site is where ASC's three inline statements would be:

```
low, high = V.vunzip(x, "uint16")
one = V.vbrc(T.uint16(1), size=lanes)
return V.vinterpret_cast(V.vor(high, V.vmin(low, one)), "bfloat16")
```

```
odd = to_bf16_round_odd(q, PAIR)
V.vstore(V.vcvt(odd, "float4_e2m1fn", rounding="R"), q_ub[col])
```

D1 per_token/04: SIMT encodes e2m1 in software

So the SIMT kernel declares its output `uint8` and extracts the float32 fields by hand — the first 6 lines of an 18-line encoder covering sign, exponent, subnormal renormalization, guard/round/sticky and saturation:

```
bits = T.reinterpret("int32", v)
# bit 3 of (bits >> 28) is bit 31 of bits: the sign nibble, without
# depending on whether >> sign-extends.
sign_nib = (bits >> 28) & 0x8
exps = (bits >> 23) & 0xFF
mant = bits & 0x7FFFFF
```

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
per_token/04	6,160	4,784 (0.78x)	613,632 (99.62x)	34 / 23 / 15

The result is bit-exact against the PyTorch reference packer. `per_block/03` carries the same encoder at 16.77x.

D1 `per_token/04`: 99.6x decomposes into instructions and divergence

factor	SIMT `per_token/04`	other SIMT `per_token` variants
instructions issued	124,294	~37,000
cycles per instruction	4.94	1.90-2.01

The second factor is branch divergence. `T.if_then_else` on a data-dependent per-element condition lowers to a branch rather than a predicated select, and this is the only variant whose trace carries a large number of `SIMT_START_DVG` markers.

Rewriting the encoder's three conditionals as mask arithmetic (`m = -T.Cast("int32", cond)`, then `(a & m) | (b & ~m)`) issues **45% more instructions** and runs at **0.56x the cycles** — 345,189 against 613,632, or **56.0x ASC instead of 99.6x**. Both spellings are in the kernel behind a `branch_free` flag and the harness asserts their outputs are byte-identical.

> ***D1*** *— an instruction the frontend cannot name is not a style difference. Of the 99.6x, a factor of 1.78x is a code-generation gap a GPU compiler would not have, since it predicates such a conditional into a select; the remaining 56.0x is a software encoder standing in for one convert instruction.*

Measured data (1/2): `cast_back`, `per_token`

variant	ops A/P	lines A/P/S	cyc A	cyc P	cyc S	P/A	S/A
`cast_back/01`	9/7	16/12/6	2,105	2,201	6,908	1.05	3.28
`cast_back/02`	12/12	12/16/9	2,201	2,233	7,033	1.01	3.20
`cast_back/03`	8/6	9/10/6	2,080	2,144	6,969	1.03	3.35
`cast_back/04`	6/6	6/7/6	2,016	2,048	6,983	1.02	3.46
`cast_back/05`	16/8	18/11/13	2,296	2,584	10,152	1.13	4.42
`cast_back/06`	14/14	21/22/16	2,272	2,528	9,275	1.11	4.08
`per_token/01`	39/19	37/24/15	6,176	4,576	71,606	0.74	11.59
`per_token/02`	45/14	43/20/17	4,608	4,688	71,008	1.02	15.41
`per_token/03`	44/15	37/21/15	4,608	4,672	74,136	1.01	16.09
`per_token/04`	37/17	34/23/15	6,160	4,784	613,632	0.78	99.62
`per_token/05`	56/38	54/48/15	5,477	5,448	75,281	0.99	13.74
`per_token/06`	39/30	57/44/32	6,864	4,576	71,606	0.67	10.43
`per_token/07`	31/17	39/24/15	4,864	5,136	72,490	1.06	14.90

`/06 full` emits `/01`'s kernel, hence the identical SIMT cycle count.

Measured data (2/2): `per_block`, `per_channel`

variant	ops A/P	lines A/P/S	cyc A	cyc P	cyc S	P/A	S/A
`per_block/01`	21/22	21/25/18	1,086	924	19,276	0.85	17.75
`per_block/02`	27/30	28/33/20	918	936	19,444	1.02	21.18
`per_block/03`	28/26	29/29/19	1,625	1,040	27,256	0.64	16.77
`per_block/04`	21/22	21/25/18	1,414	924	19,272	0.65	13.63
`per_block/05`	29/31	33/37/28	918	940	19,444	1.02	21.18
`per_channel/01`	17/16	21/21/12	502	387	2,677	0.77	5.33
`per_channel/02`	28/29	35/40/17	849	811	5,964	0.96	7.02
`per_channel/03`	23/19	29/35/14	930	721	4,073	0.78	4.38
`per_channel/04`	32/33	46/51/17	738	736	5,964	1.00	8.08

`per_block` reduces a 32x32 tile to one scalar; `per_channel` reduces along the token axis. `per_block/02` and `/05`, and `per_channel/02` and `/04`, share SIMT cycle counts for the same reason: the composed variant emits the earlier variant's `T.SimtVF` region unchanged.

Totals across the 22 variants

	PTO vs ASC	SIMT vs ASC
vector operations	74%	not comparable
source lines	89%	**53%**
simulated cycles	**0.92x** (0.64x-1.13x)	10.44x (3.20x-21.18x)

SIMT has no comparable operation count: its arithmetic is plain operators, so the backend selects the instructions.

- PTO uses fewer cycles on 9 of 22 variants, more on 6, within 2% on 7
- The SIMT range excludes `per_token/04`; including it the maximum is 99.62x
- SIMT costs under 6x on 8 of 22 variants and over 10x on 11

Caveats

1. **Simulated cycles, not hardware.** `msprof op simulator`, SoC `Ascend950PR_9599`. No silicon measurement is claimed.
2. **One shape per variant**, mostly `M=32`, `K=128`. Ratios compare three implementations of one variant and are valid; absolute cycle counts are not comparable between variants.
3. **Single core.** No multi-core scheduling, double buffering or cache hints.
4. **The SIMT kernels are not tuned.** They use the first formulation a thread-model implementation suggests. A separate SIMT-vs-SIMD suite sweeping several strategies per kernel measured 0.33x-53x, including SIMT faster on some grouped reductions.
5. **Line counts exclude helpers defined outside the VF region**, in all frontends. This understates SIMT by 18 lines on each of the two FP4 variants.

Caveat 6: the declared SIMT thread count

Every SIMT kernel declares `T.SimtVF(threads=128)`. At `K=128, group_size=32` the two passes inside the region have different extents:

```
for g in T.Parallel(num_groups):     extent 4    (the reduction)
for i in T.Parallel(hidden):         extent 128  (the apply)
```

Measured on `per_token/01`, the same kernel built at three declared widths (`harness/probe/simt_thread_count.py`, all three verified against the oracle):

threads	simulated cycles	vs 128	correct
32	71,278	1.00x	yes
64	69,047	0.96x	yes
128	71,606	1.00x	yes

Lowering the declared thread count does not recover the cost. Across a 4x range the spread is under 4%, so the penalty is not caused by declaring more threads than the reduction uses. It is the per-element execution model: ~37,000 instructions at 1.95 cycles each, against 32 VF launches of vector work.

Conclusions

SIMT reduces source size where the SIMD frontends move data between lanes. The three cases measured here are a group broadcast (B1, 16 lines to 6), a lane interleave (C1, 35 to 17) and a register gather (C2, 54 to 15). In each, the SIMD code is dominated by index arithmetic that the thread model does not require, and in `per_channel/02` SIMT also removes the 256-lane padding requirement. The cost is 3.28x, 7.02x and 13.74x simulated cycles respectively.

PTO simplifies where a segment count or a width replaces repetition (A2, B2: 39 operations to 19 and 45 to 14) **and costs source length where there is nothing to factor out** (B3: 4 lines longer at an identical operation count).

Neither SIMD frontend has anything to simplify where the operation already maps onto the hardware. When a channel occupies one lane (A1) or the value being placed is a scalar (`per_block/01`), ASC and PTO use 17 and 16 operations and 21 lines each, and SIMT's shorter source expresses the same three stages.

Where the hardware has an instruction, the frontend that can name it wins by a wide margin. All three run on the same unit; only ASC and PTO have a call that reaches the e2m1 convert. FP4 output (D1) costs SIMT 99.6x. A factor of 1.78x of that is a code-generation gap — branch divergence the mask-arithmetic spelling removes — and the remaining 56.0x is a software encoder standing in for one convert instruction.

Frontend selection

SIMT — when the kernel is dominated by data placement (packing, transposes, gathers) and development time is worth more than cycles. It is never faster here: the shorter source is bought with cycles. *per_token/05 54 lines to 15 at 13.74x · per_channel/02 35 to 17 at 7.02x · cast_back/01 16 to 6 at 3.28x*

PTO/VMI — general production vector code, and anywhere a segment count or a width replaces repetition: 74% of ASC's operations at 0.92x of its cycles. *per_token/01 39 operations to 19 at 0.74x · per_token/02 45 to 14 at 1.02x · per_block/03 at 0.64x*

ASC — when an instruction VMI does not expose is required (per-lane variable right shift, which SIMT can also express — B3), when a distribution mode is the objective, or when there is nothing for VMI to factor out. *cast_back/02 12 lines against PTO's 16 · per_channel/04 46 against 51 · cast_back/05, PTO's worst cycle regression, 1.13x*

On methodology. Source size correlates with simulated cycles at r = -0.09 on this set: a frontend comparison reporting only source size reports one of the two quantities that matter.